

Module 2

LTI Systems

Solutions

Final Answers

Question	Final Answer
Q2(a)	LTI
Q2(b)	Nonlinear (Not LTI)
Q2(c)	Time-varying (Not LTI)
Q2(d)	LTI
Q3(a)	Stable
Q3(b)	Stable only for bounded inputs; generally nonlinear , stability depends on the input set
Q3(c)	Unstable
Q3(d)	Stable
Q4(a)	Invertible; $x(t)=y(t)/2$
Q4(b)	Not invertible
Q4(c)	Invertible; $x(t) = y(t+1)$
Q4(d)	Invertible; $x[n] = y[n]-3$
Q6	{1, 3, 3, 1}
Q7	$y(t)=(1 - e^{-t}) u(t)$
Q8	{2, 5, 4, 1}

Q9(a)	Stable
Q9(b)	Unstable
Q9(c)	Stable
Q10	$x*h = h*x = \{3, 7, 2\}$